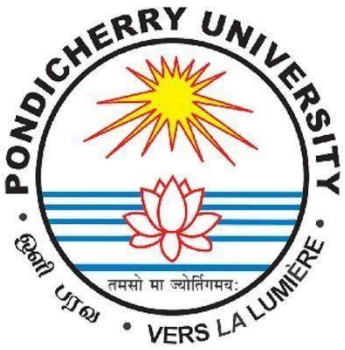
PONDICHERRY UNIVERSITY

(A Central University)



SCHOOL OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE
MASTER OF COMPUTER APPLICATIONS

CSCA 513: MINI PROJECT

NOVEMBER 2026

NAME	**:**	**SIDDHARTHA SAHA**
REG. NO	**:**	**25MCA00PY0030**
TITLE OF THE PROJECT	**:**	**VAULTGUARD: A ZERO-KNOWLEDGE PASSWORD MANAGER WITH CLIENT-SIDE ENCRYPTION**
GUIDE NAME	**:**	**DR. V. UMA**
YEAR / SEMESTER	**:**	**II / III**

BONAFIDE CERTIFICATE

This is to certify that this project work entitled "**VAULTGUARD: A ZERO-KNOWLEDGE PASSWORD MANAGER WITH CLIENT-SIDE ENCRYPTION**" is a bonafide record of work done by **MR. SIDDHARTHA SAHA** (Reg. Number **25MCA00PY0030**) in partial fulfilment for the Degree of Master of Computer Applications in the Department of Computer Science, School of Engineering and Technology of Pondicherry University.

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APPENDIX-A: SOURCE CODE

```
if (showLogin) {
    showLogin.addEventListener('click', (e) => {
        e.preventDefault();
        registerForm.style.display = 'none';
        loginForm.style.display = 'block';
        loginError.textContent = '';
        registerError.textContent = '';
    });
}

if (registerBtn) {
    registerBtn.addEventListener('click', async () => {
        const email = registerEmail.value.trim().toLowerCase();
        const password = registerPassword.value;
        registerError.textContent = '';

        if (!email || !password) {
            registerError.textContent = '⚠️ Email and password are required.';
            return;
        }
        if (password.length < 8) {
            registerError.textContent = '⚠️ Password must be at least 8 characters.';
            return;
        }

        registerBtn.disabled = true;
        registerBtn.textContent = '⏳ Creating...';

        const result = await register(email, password);

        registerBtn.disabled = false;
        registerBtn.textContent = 'Register';
```

```
        if (result.status === 'success') {
            // Generate RSA keys for this account on this device
            await generateRSAKeyPair(email);
            alert('✅ Account created! You can now log in.');
            registerForm.style.display = 'none';
            loginForm.style.display = 'block';
            loginEmail.value = email;
            loginPassword.value = '';
        } else {
            registerError.textContent = '❌ ' + (result.message || 'Registration failed.');
        }
    });
}

if (loginBtn) {
    loginBtn.addEventListener('click', async () => {
        const email = loginEmail.value.trim().toLowerCase();
        const password = loginPassword.value;
        loginError.textContent = '';

        if (!email || !password) {
            loginError.textContent = '⚠️ Email and password are required.';
            return;
        }

        loginBtn.disabled = true;
        loginBtn.textContent = '⏳ Logging in...';

        const result = await login(email, password);

        loginBtn.disabled = false;
        loginBtn.textContent = 'Login';

        if (result.status === 'success') {
```

```
            currentToken = result.token;
            currentUser = result.user;
            masterPassword = password; // used locally for vault decryption

            // Ensure RSA keys exist for sharing / switch on this device
            await generateRSAKeyPair(currentUser.email);

            showVault();
            await loadPasswords();

            // [FIX] Bootstrap an empty encrypted vault on first login so the
            // demo always has a ciphertext row in db.vaults. app.py now returns
            // status:'not_found' (HTTP 404) from GET /api/vault when no row
            // exists; we react to that by writing an encrypted [] via the real
            // PBKDF2 → AES-GCM path, so the narration stays honest.
            const existing = await getVault();
            if (existing.status === 'not_found') {
                await saveAndReload([]);
            }

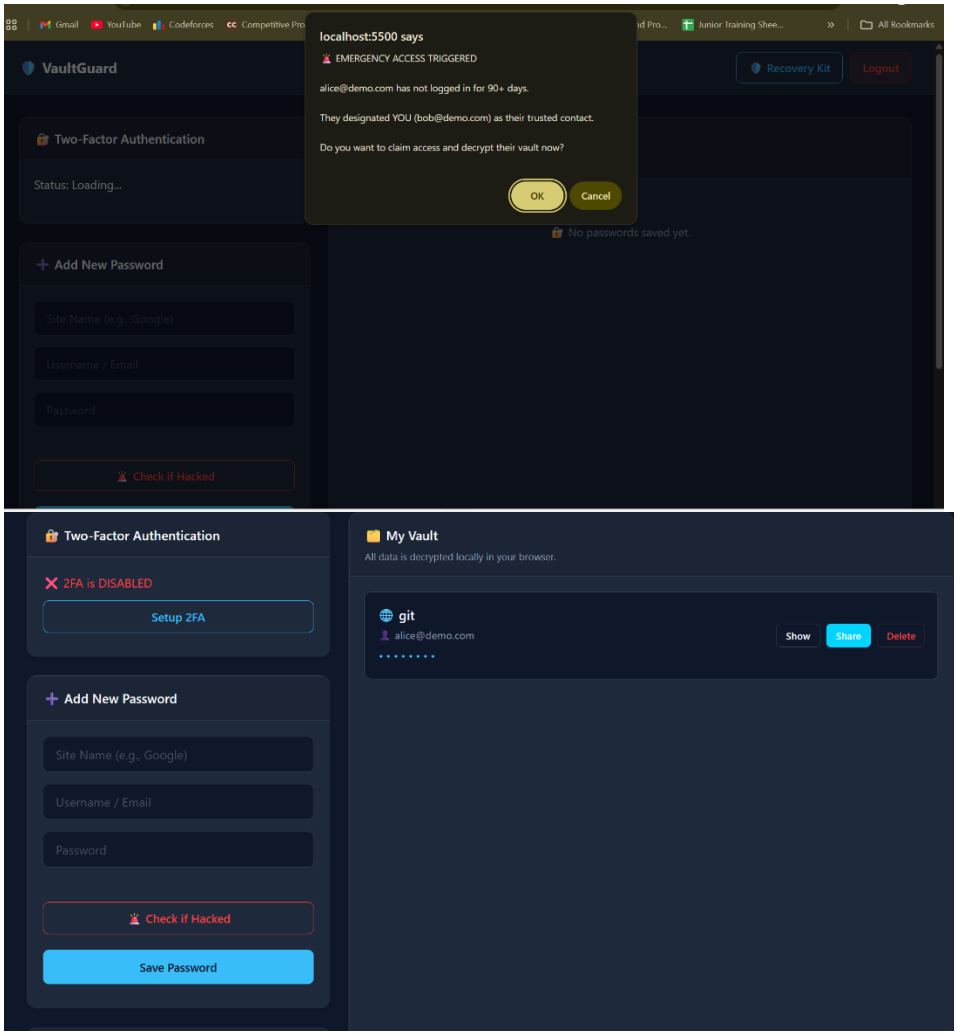
            // Check once immediately, then start polling
            await checkTriggeredSwitches();
            startSwitchPolling();
        } else {
            loginError.textContent = '❌ ' + (result.message || 'Login failed.');
        }
    });
}

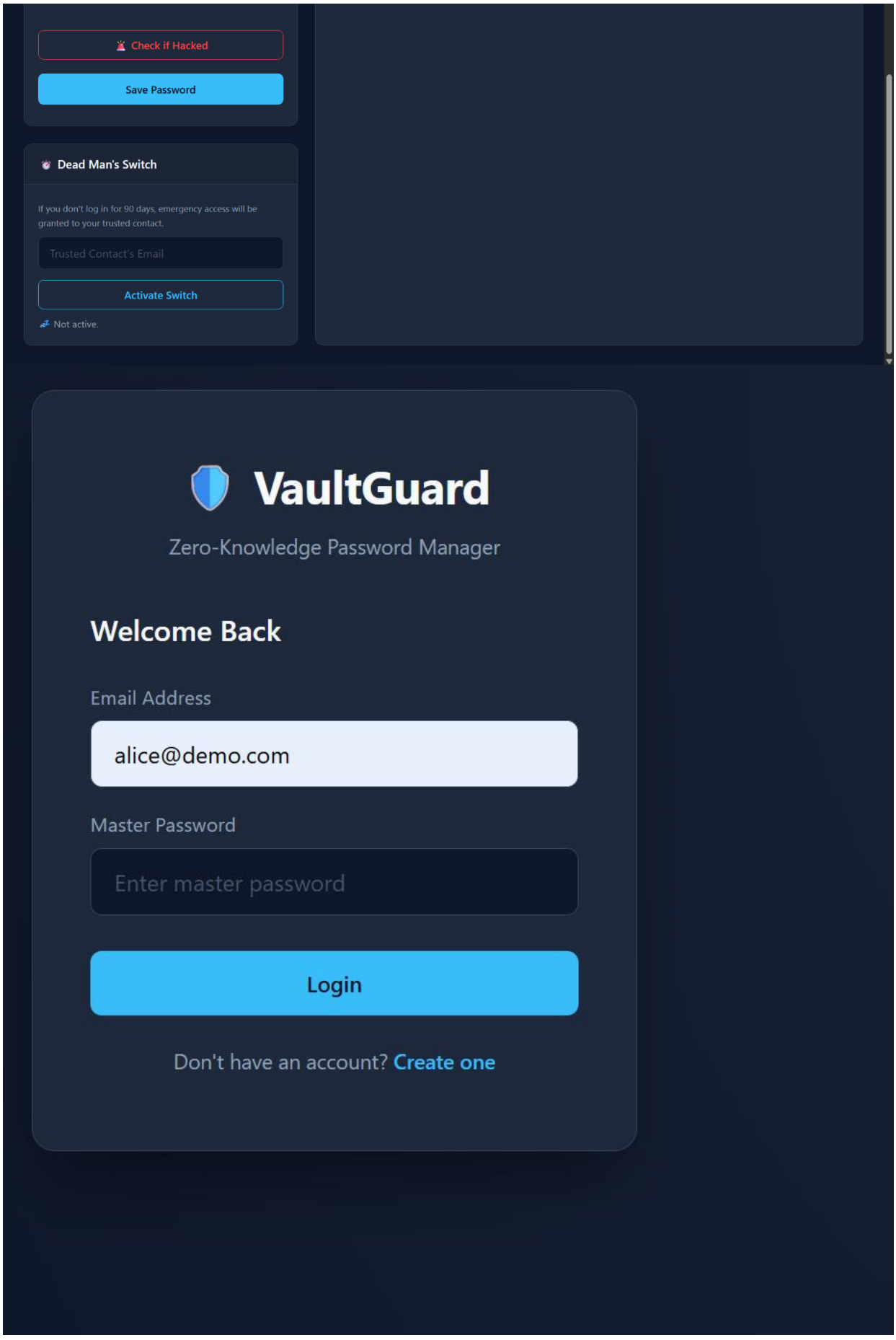
if (logoutBtn) {
    logoutBtn.addEventListener('click', () => {
        currentToken = null;
        currentUser = null;
        masterPassword = null;
```

```
        passwordsContainer.innerHTML = '';
        loginPassword.value = '';
        showAuth();
    });
}
```

APPENDIX-B: DATABASE DOCUMENTS AND PAYLOAD FORMAT

I have attached screenshots of my program below




Check if Hacked
Save Password
Dead Man's Switch
If you don't log in for 90 days, emergency access will be granted to your trusted contact.
Trusted Contact's Email
Activate Switch
Not active.
VaultGuard
Zero-Knowledge Password Manager
Welcome Back
Email Address
alice@demo.com
Master Password
Enter master password
Login
Don't have an account? Create one

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